

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

PART OF SPEECH TAGS

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- **Part-of-Speech Tags**
- POS Tagging using spaCy
- Classic POS Tagger using HMM
- More POS Tagging Ideas & Summary

Parts-of-speech are syntactic classes of words

- Not semantic

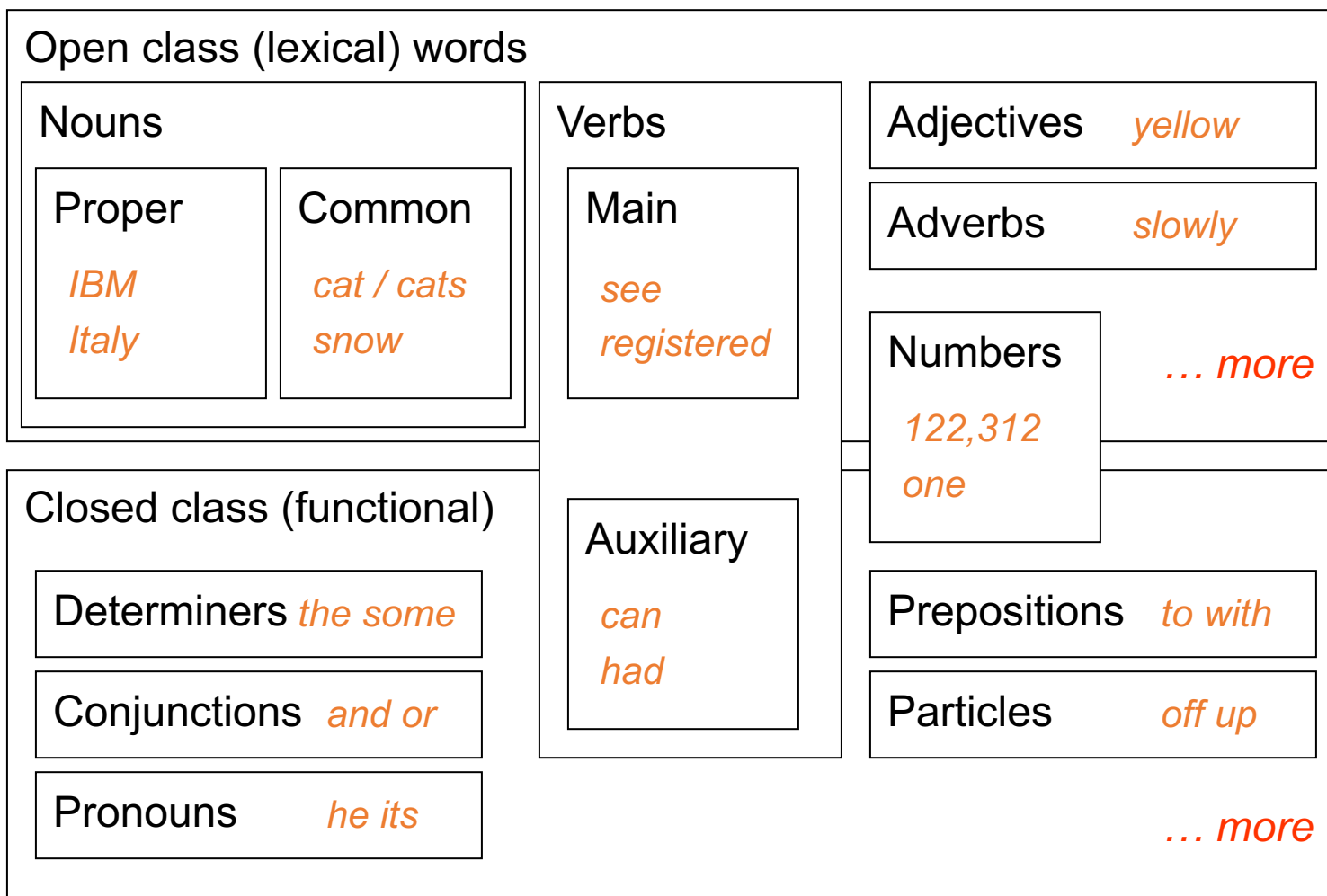
Examples:

- Coarse-grain: Noun, Verb, ...
- Fine-grain: Proper Noun, Pronoun, ...

The tags vary across languages

- Think about those language-specific phenomena

POS TAGS FOR ENGLISH



POS TAGS FOR ENGLISH

CC	conjunction, coordinating	and both but either or
CD	numeral, cardinal	mid-1890 nine-thirty 0.5 one
DT	determiner	a all an every no that the
EX	existential there	there
FW	foreign word	gemeinschaft hund ich jeux
IN	preposition or conjunction, subordinating	among whether out on by if
JJ	adjective or numeral, ordinal	third ill-mannered regrettable
JJR	adjective, comparative	braver cheaper taller
JJS	adjective, superlative	bravest cheapest tallest
MD	modal auxiliary	can may might will would
NN	noun, common, singular or mass	cabbage thermostat investment subhumanity
NNP	noun, proper, singular	Motown Cougar Yvette Liverpool
NNPS	noun, proper, plural	Americans Materials States
NNS	noun, common, plural	undergraduates bric-a-brac averages
POS	genitive marker	' 's
PRP	pronoun, personal	hers himself it we them
PRP\$	pronoun, possessive	her his mine my our ours their thy your
RB	adverb	occasionally maddeningly adventurously
RBR	adverb, comparative	further gloomier heavier less-perfectly
RBS	adverb, superlative	best biggest nearest worst
RP	particle	aboard away back by on open through
TO	"to" as preposition or infinitive marker	to
UH	interjection	huh howdy uh whammo shucks heck
VB	verb, base form	ask bring fire see take
VBD	verb, past tense	pleaded swiped registered saw
VBG	verb, present participle or gerund	stirring focusing approaching erasing
VBN	verb, past participle	dilapidated imitated reunified unsettled
VBP	verb, present tense, not 3rd person singular	twist appear comprise mold postpone
VBZ	verb, present tense, 3rd person singular	bases reconstructs marks uses
WDT	WH-determiner	that what whatever which whichever
WP	WH-pronoun	that what whatever which who whom
WP\$	WH-pronoun, possessive	whose
WRB	Wh-adverb	however whenever where why

The same word can have multiple POS tags

- It depends on contexts

VBD		VB				
VBN	VBZ	VBP	VBZ			
NNP	NNS	NN	NNS	CD	NN	
Fed raises interest rates 0.5 percent						

Which combination makes the most sense? We have to consider ...

- Grammar
- Suffixes
- Capitalization
- Gazetteers (name databases)
- ...

Useful in many applications

- Text-to-speech
- Lemmatization (we learned this in pre-processing)
- Noun-Phrase Chunking
- Regex like: $\{JJ \mid NN\}^* \{NN \mid NNS\}$

Useful as a pre-processing step for parsing